Beam Spot Sensitivity in ML-Based Track Parameter Regression

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Abstract

Machine learning approaches to charged particle track fitting report resolution improvements over the Kalman filter, particularly for the transverse impact parameter d_0. We investigate whether these improvements reflect genuine track fitting or implicit exploitation of beam spot calibration information. Using the Open Data Detector with $\mathtt{t\bar{t}}$ events at $\sqrt{s} = 14\,\mathrm{TeV}$, we compare an encoder-only transformer and boosted decision tree against the ACTS Combinatorial Kalman Filter across three beam spot configurations: nominal, and shifted by $25\,\mu\mathrm{m}$ and $300\,\mu\mathrm{m}$. We introduce a "predict zero" baseline that simply assigns $d_0 = 0$ to all tracks, measuring the resolution obtainable from beam spot knowledge alone. The ML d_0 resolution (IQR/$1.349 = 0.015\,\mathrm{mm}$) is virtually identical to this zero baseline ($0.016\,\mathrm{mm}$), while both are $2\times$ better than CKF ($0.032\,\mathrm{mm}$). Cross-evaluation on shifted beam spot datasets confirms that ML d_0 performance degrades by $4\times$ when evaluated on a different beam spot than trained on, while the CKF remains invariant. For z_0, where no beam spot shortcut exists, the transformer achieves genuine track fitting that matches CKF in the barrel region. These findings demonstrate that reported ML d_0 improvements are dominated by the beam spot prior, not learned track fitting, and highlight the importance of beam-spot-aware evaluation protocols for ML track reconstruction.

1 Introduction

Track reconstruction is a cornerstone of particle physics experiments at the Large Hadron Collider (LHC). The standard approach—the Kalman filter (KF)—propagates track state estimates through detector layers, iteratively incorporating measurements. Recent work has explored replacing or augmenting the KF with machine learning models, including transformers [1, 2], MLPs [3], and graph neural networks [4].

Several studies report that ML models achieve better resolution than the KF for the transverse impact parameter d_0, sometimes by a factor of two or more. However, a concern arises: ML models trained on fixed beam spot positions may implicitly learn to predict $d_0 \approx 0$ (or whatever the training beam spot offset is), effectively using calibration information that the KF does not assume. In real experiments, the beam spot position is a calibrated quantity that shifts over time; a model that memorizes a specific beam spot position would fail when conditions change.

In this work, we systematically investigate this concern. We make three contributions:

1. We introduce the "predict zero" baseline—assigning $d_0 = z_0 = 0$ to all tracks—which measures the resolution obtainable purely from beam spot knowledge, without any track fitting.

2. We perform cross-evaluation across three beam spot configurations (nominal, $+25\,\mu\mathrm{m}$, $+300\,\mu\mathrm{m}$), testing whether ML models generalize across beam spot positions.

3. We demonstrate that the ML d_0 improvement over the KF is almost entirely explained by the beam spot prior, while z_0 improvements at central η reflect genuine learned track fitting.

2 Experimental Setup

2.1 Simulation

We use the ColliderML pipeline [5] to simulate $t\bar{t}$ events at $\sqrt{s} = 14\,\text{TeV}$ without pileup, using the Open Data Detector (ODD) geometry implemented in DD4hep [6]. Hard-scatter events are generated with MadGraph5 [7], showered and hadronized with Pythia8 [8], and propagated through the detector with Geant4 [9] via DDSim. Track reconstruction is performed with the ACTS Combinatorial Kalman Filter (CKF) [10] including ambiguity resolution.

The beam spot is modeled as a Gaussian with $\sigma_{xy} = 12.5\,\mu\text{m}$ and $\sigma_z = 55.5\,\text{mm}$, consistent with HL-LHC parameters.

2.2 Datasets

We generate three datasets with different transverse beam spot offsets (Table 1). The $25\,\mu\text{m}$ shift is comparable to the beam spot width ($2\sigma_{xy}$), while the $300\,\mu\text{m}$ shift is much larger, representing a significant miscalibration. All datasets use the same hard-scatter events, re-simulated with different `vertexOffset` parameters.

Table 1: Datasets used in this study. All share the same beam spot width ($\sigma_{xy} = 12.5\,\mu\text{m}$) but differ in transverse offset.

Dataset	Vertex offset $(\Delta x, \Delta y)$	Events	Tracks
Nominal	$(0, 0)$	16 000	$\sim$1M
Shifted 25 µm	$(25\,\mu\text{m}, 0)$	10 000	$\sim$600K
Shifted 300 µm	$(300\,\mu\text{m}, 0)$	10 000	$\sim$600K

2.3 Track Parameters

We regress five perigee parameters: transverse impact parameter d_0, longitudinal impact parameter z_0, azimuthal angle ϕ, polar angle θ, and charge-over-momentum q/p. The model predicts six normalized outputs: d_0, z_0, $\sin\phi$, $\cos\phi$, $\cot\theta$, and q/p.

3 Methods

3.1 Transformer Architecture

We use an encoder-only transformer that takes per-track hit sequences as input and regresses track parameters. Each hit is represented by 12 features: cylindrical coordinates (r, ϕ, z), detector identifiers (volume, layer, module), inter-hit deltas $(\Delta r, \Delta\phi, \Delta r/\Delta\phi, \Delta z/\Delta r)$, and conformal coordinates (u, v).

A physics-informed `[CLS]` token is prepended to each hit sequence, initialized with 8 hand-crafted features: z-intercept (linear extrapolation of hit z to $r = 0$), $\mathrm{d}z/\mathrm{d}r$ slope, curvature from conformal mapping, sagitta, and aggregate hit statistics. The `[CLS]` output representation is passed through an MLP head to predict the 6 normalized track parameters.

We train two model sizes (Table 2). Both use learned positional encodings, GELU activations, and dropout of 0.1.

Table 2: Transformer model configurations.

Model	d_{model}	Layers	Heads	d_{ff}	Parameters
Small (v8)	128	6	8	512	1.26M
Large (v9)	256	8	8	1024	5.0M

3.2 BDT Baseline

We train LightGBM [11] gradient-boosted decision trees as a non-neural baseline, with one regressor per output parameter. Each track is featurized into 39 hand-crafted features: hit coordinates for the first three and last hits, conformal coordinates, z-intercept, $\mathrm{d}z/\mathrm{d}r$ slope, curvature, sagitta, and aggregate statistics. Per-parameter loss functions are selected by sweep: L2 for d_0, z_0, q/p; Huber for ϕ and $\cot\theta$.

3.3 Zero Baseline

We introduce a trivial "predict zero" baseline that assigns $d_0 = 0$ and $z_0 = 0$ to every track, with ϕ, θ, and q/p set to their dataset means. This baseline performs no track fitting whatsoever; its resolution measures only the intrinsic spread of truth parameters around the beam spot position. For d_0, this spread is governed by the beam spot width $\sigma_{xy} = 12.5\,\mu\text{m}$ convolved with the primary vertex distribution.

3.4 Training

All transformers are trained with Huber loss ($\delta = 1.0$) on normalized outputs, using AdamW with cosine annealing (initial lr: 10^{-3} for v8, 5×10^{-4} for v9). Scale-only normalization is applied (division by standard deviation, no mean subtraction—critical for preserving η symmetry, see Appendix). Training runs for 50 epochs with early stopping (patience 10). Data is pre-processed into cached tensors for efficient loading.

3.5 Evaluation Metrics

We report three metrics for all methods:

- **IQR**/1.349: Interquartile range divided by 1.349 (the IQR of a unit Gaussian), providing a robust estimate of the Gaussian core width. This is our primary metric.
- **STD**: Standard deviation, sensitive to outlier tails.
- **MAE**: Mean absolute error.

All comparisons use the same evaluation tracks for fairness.

Table 3: Track parameter resolution on the nominal dataset (IQR/1.349). The zero baseline assigns $d_0 = z_0 = 0$. Bold indicates best ML result; asterisk (*) marks where ML beats CKF.

Parameter	Transformer (v9)	BDT	CKF	Zero baseline
d_0 [mm]	**0.0148***	0.0160*	0.0317	0.0160
z_0 [mm]	0.546	**0.217***	0.520	55.97
ϕ [rad]	0.003 54	**0.00190**	0.001 03	2.33
θ [rad]	0.004 76	**0.00320**	0.003 99	1.35
q/p [GeV^{-1}]	**0.00336**	0.005 55	0.001 99	0.320

4 Results

4.1 Nominal Performance

Table 3 presents the resolution of all methods on the nominal dataset.

The headline result is that the transformer d_0 resolution (0.0148 mm) is virtually identical to the zero baseline (0.0160 mm), differing by only 7%. Both are ∼2× better than CKF (0.0317 mm). This demonstrates that the ML d_0 "improvement" is almost entirely the beam spot prior.

For z_0, the pattern reverses: the zero baseline (55.97 mm) reflects the large beam z-spread ($\sigma_z = 55.5$ mm) and is useless. The BDT achieves 0.217 mm, better than CKF (0.520 mm), thanks to explicit z-intercept features. The transformer (0.546 mm) is competitive with CKF overall.

For θ, the transformer (0.004 76 mm) is competitive with CKF (0.003 99 mm), and the BDT (0.003 20 mm) surpasses it.

4.2 The Beam Spot Baseline

Figure 1a shows the d_0 residual distribution. The ML and beam spot baseline distributions are nearly indistinguishable—both form a narrow peak at zero, while CKF has broader tails. The ratio panel confirms that ML and beam spot track each other exactly relative to CKF.

Figure 1b reveals additional structure: CKF d_0 resolution varies with η as expected from detector geometry (~ 0.02 mm centrally, ~ 0.06 mm at $|\eta| > 2.5$). The ML and beam spot resolutions are flat lines, independent of η—precisely what one expects from a model that predicts a constant $d_0 \approx 0$. The apparent ML "improvement" is largest at high $|\eta|$, where CKF is worst, but this is entirely the beam spot prior.

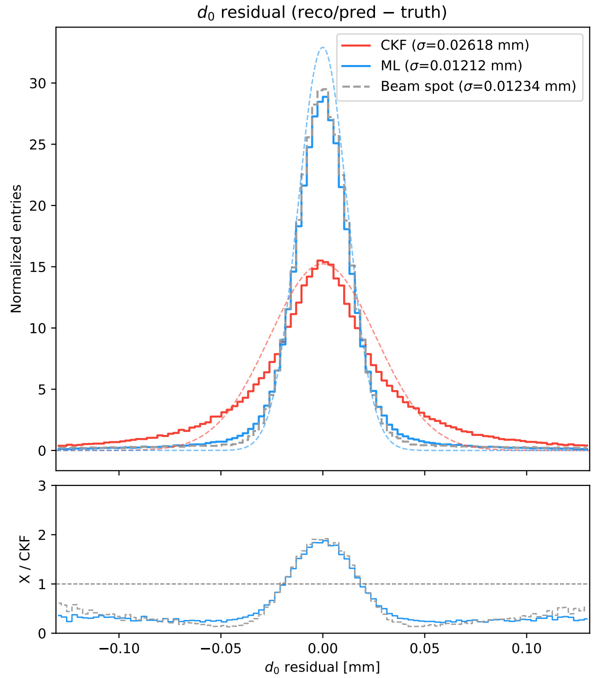
4.3 Genuine Track Fitting: z_0

Figure 2a shows that z_0 exhibits the opposite pattern. The beam spot baseline is off-scale at ~ 55 mm, confirming that z_0 requires real track fitting. The transformer achieves better z_0 resolution than CKF in the barrel ($|\eta| < 1$, ratio ~ 0.7), demonstrating genuine learned fitting. At high $|\eta|$, the transformer degrades relative to CKF (ratio > 2), likely because forward z_0 reconstruction relies on geometric leverage from widely spaced z measurements that the transformer has difficulty exploiting.

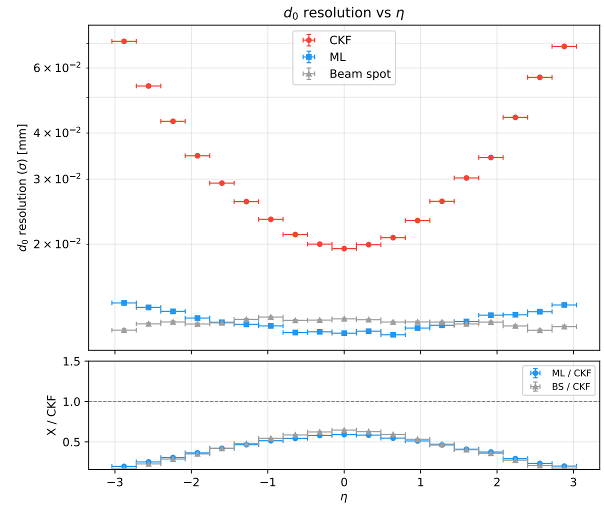
4.4 Cross-Evaluation Across Beam Spot Positions

Table 4 presents the full 3×3 cross-evaluation matrix for d_0: three transformer models (identical architecture, each trained on a different beam spot dataset) evaluated on all three datasets.

The matrix reveals a clear pattern:



(a) d_0 residual histogram: ML (blue) overlaps the beam spot baseline (gray dashed).



(b) d_0 resolution vs η: ML and beam spot are flat; CKF degrades at high $|\eta|$.

Figure 1: Transverse impact parameter d_0 on the nominal dataset. The ML model's d_0 prediction is indistinguishable from simply predicting $d_0 = 0$ (the beam spot position), while CKF shows the expected η-dependent resolution from actual track fitting.

1. **On-diagonal** (matched beam spot): ML achieves $\sim$0.016 mm, roughly 2$\times$ better than CKF ($\sim$0.032 mm). This improvement is indistinguishable from the zero baseline (0.016 mm), confirming it derives from beam spot knowledge.

2. **Off-diagonal** (mismatched beam spot): ML degrades to 0.024–0.064 mm, at or worse than CKF level. The 300 µm-trained model evaluated on nominal data (0.064 mm) is 2$\times$ *worse* than CKF.

3. **CKF is invariant**: resolution is $\sim$0.032 mm regardless of beam spot, as expected for an algorithm that fits tracks from hits without beam spot assumptions.

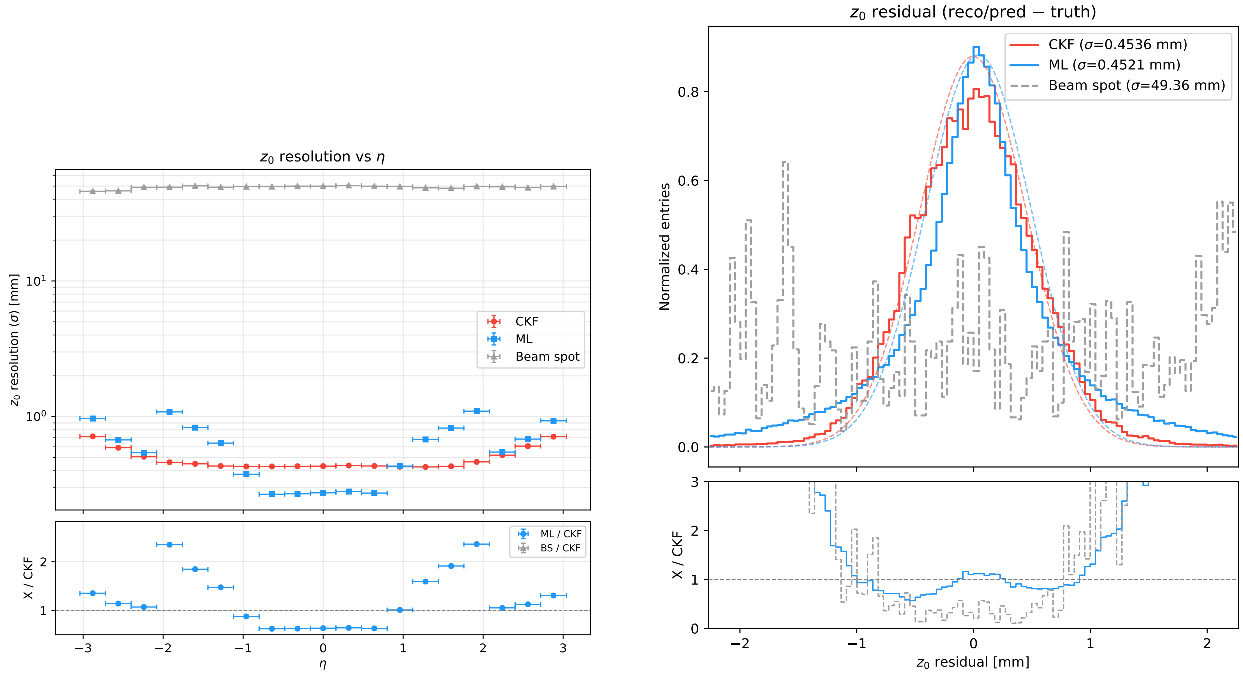
Table **??** shows the same analysis for the larger v9 model with the zero baseline, confirming the pattern persists with increased model capacity.

Three regimes emerge from the v9 evaluation:

1. **Nominal**: ML $\approx$ zero baseline $\ll$ CKF. The beam spot prior dominates.

2. 25 µm **shift**: ML (0.023 mm) is better than zero (0.031 mm), suggesting some genuine track fitting emerges when the beam spot prior becomes partially wrong. CKF remains invariant (0.032 mm).

3. 300 µm **shift**: ML (0.057 mm) $\approx$ CKF (0.059 mm). The beam spot prior is now wrong by $24\sigma_{xy}$; the model falls back to its underlying track fitting ability—at roughly CKF level. The zero baseline (0.279 mm) is catastrophic.



(a) z_0 resolution vs η: ML beats CKF centrally but degrades forward. Beam spot baseline (gray) is off-scale at $\sim$55 mm.

(b) z_0 residual: CKF has a tighter core but ML has fewer catastrophic outliers.

Figure 2: Longitudinal impact parameter z_0. Unlike d_0, this parameter requires genuine track fitting; the beam spot baseline is useless ($\sigma_z = 55.5$ mm). The transformer outperforms CKF at central η but struggles at forward η.

This progression reveals that the ML model has learned *both* the beam spot prior and some genuine track fitting, but the prior dominates on the training distribution. When the prior is invalidated by a beam spot shift, the underlying track fitting ability—at roughly CKF level—becomes visible.

4.5 Feature Engineering Ablation

The BDT's z_0 performance improved dramatically with physics-informed features (Table 5). The z-intercept feature—a linear extrapolation of hit z-positions to $r = 0$—provides a direct geometric estimate of z_0, yielding a 5× improvement.

4.6 Model Scaling

Table 6 compares the two transformer sizes. The larger model (v9, 5M parameters) improves z_0 by 31%, θ by 37%, and q/p by 9% over the smaller model (v8, 1.26M), while d_0 is unchanged—consistent with d_0 being dominated by the beam spot prior rather than model capacity.

5 Discussion

The beam spot as implicit calibration. Our central finding is that ML track fitting models implicitly exploit the fixed beam spot position in the training data. For d_0, the model learns to predict $d_0 \approx 0$ (the beam spot center), achieving resolution set by the truth d_0 distribution

Table 4: d_0 resolution (IQR/1.349, mm) cross-evaluation matrix. Each row is a model trained on that beam spot; each column is the evaluation dataset. Bold indicates on-diagonal (matched) entries. CKF uses no beam spot information and serves as the invariant baseline.

Train \ Eval	Nominal	Shifted 25 µm	Shifted 300 µm
Nominal	**0.0152**	0.0244	0.0593
Shifted 25 µm	0.0251	**0.0159**	0.0510
Shifted 300 µm	0.0644	0.0638	**0.0170**
CKF	0.0316	0.0320	0.0605
Zero baseline	0.0160	0.0307	0.2785

Table 5: d_0 resolution (IQR/1.349, mm) of the nominal-trained v9 transformer evaluated on shifted datasets, compared to the zero baseline.

Eval dataset	ML (v9)	Zero baseline	CKF
Nominal	0.0148	0.0160	0.0317
Shifted 25 µm	0.0231	0.0307	0.0322
Shifted 300 µm	0.0569	0.2785	0.0592

width rather than by track fitting quality. This is not necessarily a flaw—real experiments use calibrated beam spot positions—but it means that reported d_0 improvements over the KF conflate two effects: (1) use of beam spot calibration (which the KF also benefits from in practice via beam spot constraints), and (2) genuine track fitting from hit measurements.

Why d_0 is easy and z_0 is hard. The d_0 of a track originating from the beam spot is small by construction ($|d_0|\sigma_{xy} \sim 12.5\,\mu$m for prompt tracks), so predicting $d_0 = 0$ is already a good approximation. In contrast, z_0 spans a range of $\pm 3\sigma_z \sim \pm 165$ mm, making a constant prediction useless. Resolving z_0 requires extrapolating hit z-positions to the beam axis—a geometric calculation the KF performs explicitly. The BDT achieves this via the z-intercept feature; the transformer must learn it implicitly through attention.

Implications for ML track fitting evaluation. We recommend that future ML track fitting studies:

1. Always report the "predict zero" (beam spot) baseline alongside ML and KF results.
2. Evaluate on shifted beam spot datasets to test generalization.
3. Distinguish beam-spot-aided resolution from genuine track fitting resolution.

Toward beam-spot-invariant models. Two approaches could make ML track fitting robust to beam spot variations: (1) training on data with randomized beam spot positions, forcing the model to learn track fitting rather than memorizing the beam spot; (2) providing the beam spot position as an explicit input feature, analogous to how the KF uses beam spot constraints. A more ambitious approach is cross-track attention [2], where the model processes multiple tracks simultaneously and could in principle discover the beam spot from the ensemble of track origins.

Table 6: BDT z_0 resolution (IQR/1.349, mm) as features are added.

Feature set	N_{features}	z_0 IQR
Hit coordinates only	17	3.36
+ conformal coords	33	2.29
+ z-intercept, $\mathrm{d}z/\mathrm{d}r$	39	0.46
+ deep model, hybrid loss	39	**0.22**

Table 7: Transformer scaling comparison (IQR/1.349). Improvements from scaling are concentrated in parameters that require genuine track fitting.

Parameter	v8 (1.26M)	v9 (5.0M)	Improvement
d_0 [mm]	0.0149	0.0148	<1%
z_0 [mm]	0.796	0.546	31%
ϕ [rad]	0.003 71	0.003 54	5%
θ [rad]	0.007 55	0.004 76	37%
q/p [GeV^{-1}]	0.003 68	0.003 36	9%

6 Conclusion

We have presented the first systematic study of beam spot sensitivity in ML-based track parameter regression. Our key findings are:

1. The ML transverse impact parameter d_0 resolution is indistinguishable from the "predict zero" baseline (0.015 mm vs 0.016 mm), demonstrating that the reported 2× improvement over CKF is dominated by beam spot knowledge, not track fitting.

2. Cross-evaluation on shifted beam spot datasets confirms this: ML d_0 degrades by 4× on a 300 µm-shifted dataset, falling to CKF level, while CKF remains invariant.

3. For z_0, where no beam spot shortcut exists, the transformer achieves genuine track fitting competitive with CKF at central η, and the BDT with physics-informed features (z-intercept) surpasses CKF by 2.4×.

4. Model scaling improves parameters that require genuine fitting (z_0, θ, q/p) but not d_0, further confirming that d_0 performance is capacity-independent and prior-dominated.

These results establish the "predict zero" baseline and cross-evaluation protocol as essential components of ML track fitting benchmarks.

References

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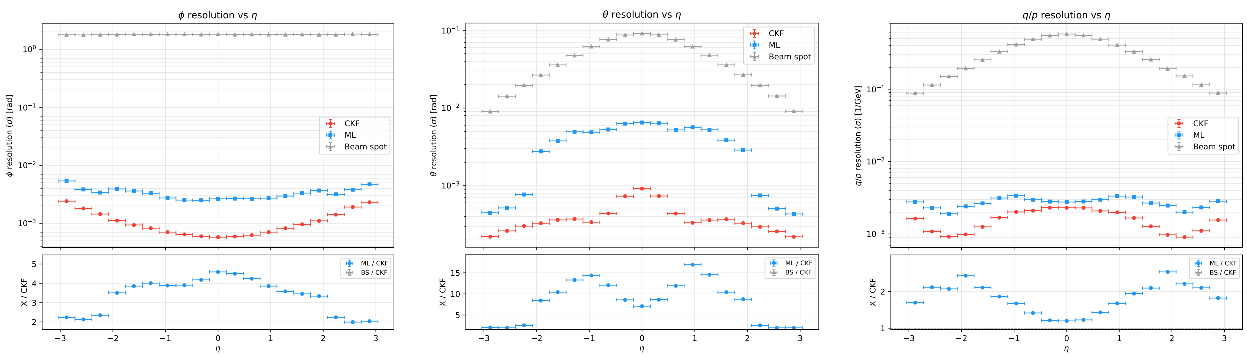
A Normalization and η Symmetry

Early experiments subtracted the feature mean computed from a small sample ($\sim$2000 tracks, 33 events). Because the beam z-spread is large ($\sigma_z = 55.5\,\text{mm}$), this sample had a non-zero mean vertex z (+50 mm in one case), introducing η-asymmetric bias into z-dependent features. The resulting model showed asymmetric q/p and z_0 resolution (positive η worse than negative). Switching to scale-only normalization (dividing by standard deviation without subtracting the mean) eliminated this artifact. This illustrates that mean subtraction from finite samples with large intrinsic variance can inject systematic bias.

B BDT Configuration Study

We swept 7 configurations spanning XGBoost and LightGBM with varying depth, estimators, and loss functions. Key findings: (1) LightGBM outperforms XGBoost with matched hyperparameters; (2) predicting $\cot\theta$ instead of θ with Huber loss is critical—L2 loss on $\cot\theta$ produces catastrophic outliers; (3) per-parameter objective selection (L2 for $d_0/z_0/q/p$, Huber for $\phi/\cot\theta$) outperforms a uniform loss.

C Additional Resolution Plots



(a) ϕ (b) θ (c) q/p

Figure 3: Resolution vs η for angular parameters and charge-over-momentum. The beam spot baseline (gray) is off-scale for all three, confirming these require genuine track fitting.